DELETION FORMULAS FOR CHOW FUNCTIONS

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Abstract.

1. Introduction

1.1. **Main results.**

2. Preliminaries

2.1. **Posets and Incidence Algebras.** Let $\mathcal{P}$ be a *locally finite, weakly ranked, partially ordered set*. We denote by $\mathrm{Int}(\mathcal{P})$ the set of all closed intervals of $\mathcal{P}$ and by $\rho : \mathrm{Int}(\mathcal{P}) \to \mathbb{Z}$ the *weak rank function* satisfying $\rho_{st} = \rho_{sw} + \rho_{wt}$ for any $s \leq w \leq t$.

The *incidence algebra* of $\mathcal{P}$ over $\mathbb{Z}[x]$, denoted $\mathcal{I}$, is the free $\mathbb{Z}[x]$-module on $\mathrm{Int}(\mathcal{P})$. An element $a \in \mathcal{I}$ is a function that assigns a polynomial $a_{st}(x)$ to each interval $[s,t]$. The algebra structure is given by the convolution product:

$$(ab)_{su} := \sum_{s \leq t \leq u} a_{st}(x) \cdot b_{tu}(x), \qquad \text{for every } s \leq u \text{ in } \mathcal{P}.$$

The algebra is associative with a multiplicative identity δ. An element $a \in \mathcal{I}$ is invertible if and only if $a_{ss}(x)$ is a unit for every $s \in \mathcal{P}$. Two fundamental elements are the *zeta function*, $\zeta_{st}(x) = 1$, and its inverse, the *Möbius function*, $\mu = \zeta^{-1}$.

2.2. **Kernels and the KLS theory.** The weak rank function ρ induces a subalgebra $\mathcal{I}_\rho \subseteq \mathcal{I}$ consisting of elements a such that $\deg(a_{st}) \leq \rho_{st}$ for every interval $[s,t]$. This subalgebra is equipped with an involution $a \mapsto a^{\mathrm{rev}}$ defined by:

$$(a^{\mathrm{rev}})_{st}(x) := x^{\rho_{st}} a_{st}(x^{-1})$$

An element $\kappa \in \mathcal{I}_\rho$ is a $(\mathcal{P},\rho)$*-kernel* if $\kappa_{ss}(x) = 1$ and it satisfies the property $\kappa^{-1} = \kappa^{\mathrm{rev}}$. The primary example is the *characteristic function* $\chi := \mu \cdot \zeta^{\mathrm{rev}}$, which is a kernel for any weakly ranked poset.

As formulated in **[?]**, a kernel uniquely determines two key functions:

Theorem 2.1 *[Kazhdan-Lusztig-Stanley Functions] Let $\kappa \in \mathcal{I}_\rho$ be a $(\mathcal{P},\rho)$-kernel. There exist unique elements $f, g \in \mathcal{I}_\rho$ (the right and left KLS functions, respectively) satisfying:*

(i) $f_{ss}(x) = 1$ *and* $g_{ss}(x) = 1$ *for all* $s \in \mathcal{P}$.
(ii) $\deg f_{st}(x) < \frac{1}{2}\rho_{st}$ *and* $\deg g_{st}(x) < \frac{1}{2}\rho_{st}$ *for all* $s < t$.
(iii) $f^{\mathrm{rev}} = \kappa \cdot f$ *and* $g^{\mathrm{rev}} = g \cdot \kappa$.

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2.3. **Chow Functions and Related Invariants.** From a kernel κ and its KLS functions, one can define several other important polynomial invariants. Central to this paper is the *Chow function*, which is formulated in [**?**] as follows:

Definition 2.2 [Reduced Kernel and Chow Function] Let κ be a $(\mathcal{P}, \rho)$-kernel.

(1) The *reduced kernel* $\overline{\kappa} \in \mathcal{I}_\rho$ is given by:

$$\overline{\kappa}_{st}(x) = \begin{cases} \frac{\kappa_{st}(x)}{x-1} & \text{if } s < t, \\ -1 & \text{if } s = t. \end{cases}$$

(2) The *κ-Chow function* $\underline{\mathrm{H}} \in \mathcal{I}_\rho$ is defined as:

$$\underline{\mathrm{H}} = -(\overline{\kappa})^{-1}$$

The theory also provides three related symmetric functions.

Definition 2.3 [Symmetric Invariants] Let κ be a $(\mathcal{P}, \rho)$-kernel with KLS functions f and g, and Chow function $\underline{\mathrm{H}}$.

(1) The *Z-function* is $Z := g^{\mathrm{rev}} \cdot f = g \cdot f^{\mathrm{rev}}$.
(2) The *right augmented Chow function* is $F := \underline{\mathrm{H}} \cdot f^{\mathrm{rev}}$.
(3) The *left augmented Chow function* is $G := g^{\mathrm{rev}} \cdot \underline{\mathrm{H}}$.

These functions satisfy fundamental symmetry properties that we will use throughout this work. For any interval $[s, t]$ with $s < t$:

- $\underline{\mathrm{H}}_{st}(x) = x^{\rho_{st}-1}\underline{\mathrm{H}}_{st}(x^{-1})$.
- $Z_{st}(x) = x^{\rho_{st}} Z_{st}(x^{-1})$.
- $F_{st}(x) = x^{\rho_{st}} F_{st}(x^{-1})$ and $G_{st}(x) = x^{\rho_{st}} G_{st}(x^{-1})$.

3. The Module of Poset Elements and Perverse Subgroups

Let $R := \mathbb{Z}[x, x^{-1}]$ be the ring of Laurent polynomials with integer coefficients. For any integer n, we denote by $\mathrm{Pal}(n)$ the additive group of polynomials $f(x) \in R$ satisfying $f(x) = x^n f(x^{-1})$. While $\mathrm{Pal}(0)$ is a subring of R, we note that $\mathrm{Pal}(n)$ for $n \neq 0$ is not, as $\mathrm{Pal}(n) \cdot \mathrm{Pal}(m) \subseteq \mathrm{Pal}(n+m)$.

Let $\mathcal{P}$ be a locally finite, weakly ranked poset. We assume $\mathcal{P}$ to be bounded with the minimum element $\hat{0}$ and the maximum element $\hat{1}$, which allows for the definition of a global rank function $\rho(t) := \rho_{\hat{0},t}$. Let κ be a $(\mathcal{P}, \rho)$-kernel with left KLS function g.

Definition 3.1 The module $\mathcal{H}(\mathcal{P})$ is the free R-module with a standard basis $\{\mathrm{s}\}_{s \in \mathcal{P}}$. An element $\alpha \in \mathcal{H}(\mathcal{P})$ is a formal sum

$$\alpha = \sum_{s \in \mathcal{P}} \alpha_s \cdot \mathrm{s}, \quad \text{where } \alpha_s \in R.$$

3.1. **The Perversity Condition via Convolution.** The notion of a perverse element in this work is a symmetry condition motivated by the axiomatic properties of the KLS and Chow functions in the Definition 2.3.

Definition 3.2 Let g be the left KLS function associated with a kernel κ. For any $s \in \mathcal{P}$, we define the R-linear map $S_s : \mathcal{H}(\mathcal{P}) \to R$ by

$$S_s(\alpha) := \sum_{t \geq s} g_{st}^{\mathrm{rev}}(x^{-1}) \cdot \alpha_t.$$

This definition is crafted so that for the characteristic kernel ($\kappa = \chi$), it recovers the operator $S_s(\alpha) = \sum_{t \geq s} x^{\rho(s)-\rho(t)}\alpha_t$.

In **[?]**, the authors define the subgroup of **perverse elements** $\mathcal{H}_p(\mathcal{P}) \subseteq \mathcal{H}(\mathcal{P})$. We adapt their definition to define a larger subgroup $\mathcal{H}'_p(\mathcal{P}) \subseteq \mathcal{H}(\mathcal{P})$ as follows:

Definition 3.3 The subgroup $\mathcal{H}'_p(\mathcal{P}) \subseteq \mathcal{H}(\mathcal{P})$ consists of all elements α that satisfy

$$S_s(\alpha) \in \mathrm{Pal}(0) \quad \text{for all } s \in \mathcal{P}.$$

3.2. **The Chow and Kazhdan-Lusztig Bases.** Let κ be a $(\mathcal{P}, \rho)$-kernel, with associated KLS functions f and g, and Chow function $\underline{\mathrm{H}}$. Using these functions, we have the following two important bases for the module $\mathcal{H}(\mathcal{P})$.

Definition 3.4 The **Kazhdan-Lusztig basis** $\{\mathrm{kl}^t\}_{t \in \mathcal{P}}$ is defined by

$$\mathrm{kl}^t = \sum_{s \leq t} \mathrm{kl}^t_s \cdot \mathrm{s} := \sum_{s \leq t} g_{st}^{\mathrm{rev}}(x) \cdot f_{st}(x^{-2}) \cdot \mathrm{s},$$

where f is the right KLS function of κ.

Definition 3.5 The **Chow basis** $\{\mathrm{c}^t\}_{t \in \mathcal{P}}$ is defined by

$$\mathrm{c}^t = \sum_{s \leq t} \mathrm{c}^t_s \cdot \mathrm{s} := \sum_{s \leq t} g_{st}^{\mathrm{rev}}(x) \cdot \underline{\mathrm{H}}_{st}(x^{-2}) \cdot \mathrm{s},$$

where $\underline{\mathrm{H}}$ is the κ-Chow function.

Remark 3.6 From the properties of Chow functions, the coefficients have a special structure: $\mathrm{c}^t_t = 1 \in \mathrm{Pal}(0)$, and $\mathrm{c}^t_s \in \mathrm{Pal}(2)$ for all $s < t$.

Definition 3.7 We define the module $\mathcal{S}(\mathcal{P})$ to be the free $\mathrm{Pal}(0)$-module with basis $\{\mathrm{c}^t\}_{t \in \mathcal{P}}$.

Proposition 3.8 *For all $t \in \mathcal{P}$, $\mathrm{c}^t \in \mathcal{H}'_p(\mathcal{P})$, i.e., it satisfies $S_s(\mathrm{c}^t) \in \mathrm{Pal}(0)$ for all $s \in \mathcal{P}$.*

Proof. For any $s \in \mathcal{P}$, if $s \not\leq t$, then $S_s(\mathrm{c}^t) = 0$. If $s \leq t$, we have

$$\begin{aligned} S_s(\mathrm{c}^t) &= \sum_{v : s \leq v \leq t} g_{sv}^{\mathrm{rev}}(x^{-1}) \cdot \mathrm{c}^t_v \\ &= \sum_{v : s \leq v \leq t} g_{sv}^{\mathrm{rev}}(x^{-1}) \cdot g_{vt}^{\mathrm{rev}}(x) \cdot \underline{\mathrm{H}}_{vt}(x^{-2}) \\ &=^{?} g_{st}^{\mathrm{rev}}(x) \cdot \sum_{v : s \leq v \leq t} g_{sv}^{\mathrm{rev}}(x^{-2}) \cdot \underline{\mathrm{H}}_{vt}(x^{-2}) \\ &= g_{st}^{\mathrm{rev}}(x) \left(g^{\mathrm{rev}}\underline{\mathrm{H}}\right)_{st}(x^{-2}) = g_{st}^{\mathrm{rev}}(x) G_{st}(x^{-2}) \end{aligned}$$

We have that $g_{st}^{\mathrm{rev}}(x) \in \mathrm{Pal}(2\rho_{st})$ and $G_{st}(x^{-2}) \in \mathrm{Pal}(-2\rho_{st})$. Their product is therefore in $\mathrm{Pal}(0)$, as required. □

Proposition 3.9 *The subgroup $\mathcal{H}'_p(\mathcal{P})$ is a $\mathrm{Pal}(0)$-module.*

Proof. Let $\alpha = \sum_s p_s \cdot \mathrm{s} \in \mathcal{H}'_p(\mathcal{P})$ and $q \in \mathrm{Pal}(0)$. We see that

$$S_t(q \cdot \alpha) = \sum_{v \geq t} x^{\rho(t)-\rho(v)}(q \cdot p_v) = q \cdot S_t(\alpha) \in \mathrm{Pal}(0),$$

since $\mathrm{Pal}(0)$ is a ring. Therefore, $q \cdot \alpha \in \mathcal{H}'_p(\mathcal{P})$. □

Proposition 3.10 *Any Laurent polynomial f can be uniquely written as a sum $f = f_0 + f_2$ where $f_0 \in \mathrm{Pal}(0)$ and $f_2 \in \mathrm{Pal}(2)$. The projections are given by*

$$f_0(x) = \frac{x^2 f(x^{-1}) - f(x)}{x^2 - 1} \quad \textit{and} \quad f_2(x) = \frac{f(x) - f(x^{-1})}{1 - x^{-2}}.$$

Proposition 3.11 $\mathcal{S}(\mathcal{P}) = \mathcal{H}'_p(\mathcal{P})$ *as* $\mathrm{Pal}(0)$*-modules.*

Proof. Proposition 3.8 and 3.9 show that $\mathcal{S}(\mathcal{P}) \subseteq \mathcal{H}'_p(\mathcal{P})$. We now show the reverse inclusion.

Let $\alpha = \sum_s \alpha_s \cdot \mathrm{s}$ be in $\mathcal{H}'_p(\mathcal{P})$. Since $\{\mathrm{c}^s\}$ is a basis for $\mathcal{H}(\mathcal{P})$, α has a unique expansion $\alpha = \sum_{s \in \mathcal{P}} q_s \cdot \mathrm{c}^s$ for $q_s \in R$. We must prove that each $q_s \in \mathrm{Pal}(0)$ by induction on the corank of s.

The change-of-basis formula is $\alpha_s = \sum_{t \geq s} q_t \cdot \mathrm{c}^t_s$.

Base Case: For $s = \hat{1}$ (the maximal element), this simplifies to $q_{\hat{1}} = \alpha_{\hat{1}}$. Since $\alpha \in \mathcal{H}'_p(\mathcal{P})$, we have $S_{\hat{1}}(\alpha) = \alpha_{\hat{1}} \in \mathrm{Pal}(0)$, establishing the base case.

Inductive Step: Assume for all t with $\rho(t) > \rho(s)$, we have $q_t \in \mathrm{Pal}(0)$. We know $S_s(\alpha) \in \mathrm{Pal}(0)$. We also have $S_s(\alpha) = \sum_{v \geq s} x^{\rho(s)-\rho(v)} \alpha_v$. Substituting the basis change:

$$S_s(\alpha) = \sum_{v \geq s} x^{\rho(s)-\rho(v)} \left(\sum_{t \geq v} q_t \cdot \mathrm{c}^t_v \right) = \sum_{t \geq s} q_t \left(\sum_{v : s \leq v \leq t} x^{\rho(s)-\rho(v)} \mathrm{c}^t_v \right) = \sum_{t \geq s} q_t \cdot S_s(\mathrm{c}^t).$$

Isolating the $t = s$ term gives $S_s(\alpha) = q_s \cdot S_s(\mathrm{c}^s) + \sum_{t > s} q_t \cdot S_s(\mathrm{c}^t)$. Since $S_s(\mathrm{c}^s) = 1$, we rearrange to get:

$$q_s = S_s(\alpha) - \sum_{t > s} q_t \cdot S_s(\mathrm{c}^t).$$

By the inductive hypothesis, each q_t is in $\mathrm{Pal}(0)$. By Proposition 3.8, each $S_s(\mathrm{c}^t)$ is in $\mathrm{Pal}(0)$. Since $\mathrm{Pal}(0)$ is a ring, the entire sum is in $\mathrm{Pal}(0)$. Thus, q_s must also be in $\mathrm{Pal}(0)$. □

Proposition 3.12 (Change of Basis Formula) *If $\alpha = \sum_s \alpha_s \cdot \mathrm{s} \in \mathcal{S}(\mathcal{P})$, then its expansion in the c^s basis is $\alpha = \sum_s \overline{\alpha_s} \cdot \mathrm{c}^s$, where $\overline{\alpha_s}$ is the unique $\mathrm{Pal}(0)$ component of α_s from the decomposition in Proposition 3.10.*

Proof. Let $\alpha \in \mathcal{S}(\mathcal{P})$. By definition, $\alpha = \sum_s q_s \cdot \mathrm{c}^s$ where each $q_s \in \mathrm{Pal}(0)$. The change-of-basis formula gives:

$$\alpha_s = q_s \cdot \mathrm{c}^s_s + \sum_{t > s} q_t \cdot \mathrm{c}^t_s = q_s + \sum_{t > s} q_t \cdot \mathrm{c}^t_s.$$

We proceed by induction on the corank of s. The base case for $s = \hat{1}$ is $q_{\hat{1}} = \alpha_{\hat{1}}$, and since $q_{\hat{1}} \in \mathrm{Pal}(0)$, we have $q_{\hat{1}} = \overline{\alpha_{\hat{1}}}$.

For the inductive step, assume $q_t = \overline{\alpha_t}$ for all $t > s$. For any $t > s$, $q_t \in \mathrm{Pal}(0)$ and $\mathrm{c}^t_s \in \mathrm{Pal}(2)$, so their product is in $\mathrm{Pal}(2)$. The sum $\sum_{t > s} q_t \cdot \mathrm{c}^t_s$ is therefore in $\mathrm{Pal}(2)$. The equation shows α_s is the sum of q_s (which is in $\mathrm{Pal}(0)$) and a term in $\mathrm{Pal}(2)$. By the uniqueness of the decomposition in Proposition 3.10, q_s must be the unique $\mathrm{Pal}(0)$ component of α_s. □

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